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Defect concentration analysis: Combining Laplace deep-level transient spectroscopy with constrained curve fitting

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Christoph Klein,[a)] Carmine Pellegrino, and David Lackner

AFFILIATIONS

Fraunhofer Institute for Solar Energy Systems ISE, Heidenhofstrasse 2, 79110 Freiburg, Germany

[a)]**Author to whom correspondence should be addressed:** christoph.klein@ise.fraunhofer.de

ABSTRACT

Deep-level transient spectroscopy is the standard technique for characterizing electrically active defects in semiconductors. The conventional double-boxcar evaluation fails when multiple defect emissions overlap, causing peak broadening or merging. Two established routes overcome this resolution limitation: The numerical inverse Laplace transform resolves closely spaced emission rates but inherits regularization-dependent peak shapes that bias the recovered amplitudes, particularly for imbalanced peaks, while direct multi-exponential time-domain fitting avoids regularization yet is sensitive to noise and prone to overfitting when rates and amplitudes are determined simultaneously as free parameters. This work presents a two-step evaluation routine that combines Laplace transform analysis with constrained multi-exponential curve fitting of the capacitance transient. The inverse Laplace transform first identifies the emission rates, which are then held fixed in a constrained least-squares fit that recovers the individual amplitudes from the time-domain transient, thereby combining the rate resolution of Laplace DLTS with a noise-robust, regularization-free amplitude determination. The method requires no material-specific assumptions and is applicable wherever Laplace DLTS can resolve multiple defect emission rates, providing a practical tool for quantitative defect concentration analysis in semiconductor development. Validation with synthetic data demonstrates accurate recovery of defect concentrations. Application to n-type GaAs, where the well-known *EL2* defect overlaps with the *ELO* level, reveals *ELO* at a concentration 20 times lower than that of *EL2*. Despite this difference, the order-of-magnitude higher capture cross section of *ELO* renders both defects comparably important for carrier recombination, highlighting the importance of accurate concentration determination via the presented method.

I. INTRODUCTION

Deep-level transient spectroscopy (DLTS), introduced by Lang in 1974, remains the standard technique for characterizing electrically active defects in semiconductors.[1] Since then, the technique has been widely applied to characterize electrically active defects across diverse material systems, including irradiated and pristine silicon,[2,3] III–V semiconductors including AlGaInP for multijunction architectures[4] and GaInAsP for space photovoltaics,[5] wide- and ultra-wide-bandgap semiconductors such as 4H-SiC[6] and Ga_2O_3,[7,8] thin-film and perovskite solar cells,[9,10] and two-dimensional materials,[11] while methodological advances continue to extend its analytical capabilities.[12]

The conventional double-boxcar evaluation extracts defect parameters—activation energy, capture cross section, and concentration—from the temperature-dependent capacitance transient following a voltage pulse. However, this approach requires well-separated, mono-exponential transients. When multiple defects exhibit similar emission rates, their signals overlap, causing peak broadening that prevents accurate characterization of individual traps.[13] The numerical inverse Laplace transform, introduced by Dobaczewski *et al.*, overcomes this limitation by transforming the capacitance transient into an emission rate spectrum.[14,15] This technique resolves defects with emission rate ratios as low as 2:1, compared to approximately 10:1 for the double-boxcar method when the defects exhibit comparable signal amplitudes, enabling separation of closely spaced defect levels. Laplace deep-level transient spectroscopy has since become the method of choice for precise emission-rate determination and defect identification.

For the calculation of individual defect concentrations from a resolved Laplace spectrum, local integration of the spectral density function around each emission-rate peak provides the established route to the corresponding capacitance amplitude.[14,15] This approach reconstructs the amplitude, a time-domain quantity natively defined in the capacitance transient, via the detour through the regularization-dependent emission-rate spectrum, and its accuracy therefore depends on how faithfully the spectral peak shape reflects the underlying transient. Tikhonov-regularized inversions, in particular, tend to narrow minor peaks coexisting with a much larger dominant peak and to symmetrize intrinsically asymmetric peaks,[15] biasing the integrated amplitude precisely in those regimes where multi-defect analysis is most needed. An alternative route avoids the spectral domain entirely by determining the amplitudes directly in the time domain through multi-exponential least-squares fitting of the capacitance transient, as implemented in Provencher's DISCRETE algorithm.[16] Treating emission rates and amplitudes simultaneously as free parameters, however, renders this single-step approach sensitive to noise and prone to overfitting when the model order, meaning the number of overlapping defect signals, is not known beforehand.

The two-step routine proposed in this work combines the complementary strengths of both approaches. In the first step, the inverse Laplace transform identifies the emission rates contributing to the transient, and an Arrhenius analysis yields the activation energy and apparent capture cross section of each defect. In the second step, the emission rates are held fixed at the Arrhenius-consistent values, and a constrained least-squares fit of the multi-exponential model to the time-domain transient determines the individual capacitance amplitudes. Decoupling rate identification from amplitude determination retains the high emission-rate resolution of Laplace DLTS while extracting the amplitudes in the time domain, where the defect concentration is natively defined, yielding a substantially more robust workflow than the unconstrained multi-exponential fit alone.

We validate the method using synthetic data and apply it to n-type GaAs, where the well-known *EL2* defect overlaps with the oxygen-related *ELO* level.[17] Despite differing by a factor of 20 in concentration, both defects contribute comparably to carrier recombination due to their different capture cross sections—a finding only accessible through accurate individual concentration determination. The method provides a practical tool for quantitative defect analysis in semiconductors, with direct implications for optimizing material quality, e.g., in photovoltaic applications.[18]

II. THEORY

Deep-level transient spectroscopy probes electrically active defects through their carrier emission dynamics. The applied bias voltage modulates the space charge region of a semiconductor junction, within which defect states capture and emit charge carriers. A filling pulse reduces the reverse bias, narrowing the space charge region and allowing defect states to capture charge carriers.

Upon restoration of the reverse bias, the depletion region expands and the captured carriers are thermally emitted at a characteristic rate $e(T)$ determined by the defect's activation energy and capture cross section.[13] The resulting exponential capacitance transient forms the basis of defect characterization. For a single trap state, the capacitance transient following the filling pulse is given by[13]

$$C(t, T) = C(V_{\mathrm{R}}, T) + \Delta C(T) \exp(-e(T) \cdot t). \tag{1}$$

Here, $C(V_{\mathrm{R}}, T)$ denotes the quiescent capacitance at reverse voltage V_{R}, and $\Delta C(T)$ is the capacitance deflection caused by the captured charge carriers. The sign of ΔC reflects the carrier type: Majority carrier capture compensates charge within the space charge region, causing it to expand and yielding $\Delta C < 0$, whereas minority carrier capture adds to the space charge, contracting the depletion region and yielding $\Delta C > 0$.

Assuming complete occupation of all defect states with a single charge during the filling pulse, the defect concentration follows directly from the capacitance deflection as

$$N_{\mathrm{T}} = 2N_{D,A} \frac{|\Delta C(T)|}{C(V_{\mathrm{R}}, T)} \cdot \gamma(E_{\mathrm{A}}), \tag{2}$$

where $N_{D,A}$ is the donor or acceptor concentration of the semiconductor.[13] The correction factor $\gamma(E_{\mathrm{A}})$ accounts for the fraction of the space charge region probed during the measurement and depends on the trap's energetic position within the bandgap and the applied voltage conditions.[13,19]

The conventional double-boxcar method extracts ΔC by correlating the DLTS signal amplitude with the capacitance deflection through an analytical relationship that assumes a purely mono-exponential transient.[1,13] For a single defect with a well-defined emission rate, this approach accurately recovers the concentration. However, when two or more defects exhibit similar emission rates, their transients superimpose to form a multi-exponential decay. The resulting DLTS signal appears broadened or asymmetric, preventing clear assignment of peaks to individual defects and yielding erroneous ΔC values that do not correspond to any single defect.

Such a multi-exponential transient is the linear superposition of K exponential decays, one per defect level. The multi-defect generalization of Eq. (1) reads

$$C(t, T) = C(V_{\mathrm{R}}, T) + \sum_{k=1}^{K} \Delta C_k(T) \exp(-e_k(T)t), \tag{3}$$

with ΔC_k being the capacitance deflection of defect k and e_k being its emission rate. Recovering the individual amplitudes ΔC_k from a measured transient, and thereby the individual defect concentrations $N_{\mathrm{T},k}$ via Eq. (2), is the central numerical task addressed in Sec. III.

III. METHODS

Two established approaches to recovering the individual ΔC_k from Eq. (3) beyond the conventional double-boxcar evaluation are summarized below, before the combined routine proposed in this work is introduced in Sec. III B.

A. Established methods for defect concentration determination beyond the double-boxcar approach

The DISCRETE algorithm, originally developed by Provencher,[16] addresses the multi-exponential problem entirely in the time domain. It fits Eq. (3) directly to the measured transient by non-linear least-squares minimization, treating both the emission rates $\{e_k\}$ and the capacitance deflections $\{\Delta C_k\}$ as free parameters. Including the baseline $C(V_R, T)$, the optimization thus involves $2K+1$ unknowns and is performed in a single step. To mitigate the non-convexity of the optimization landscape in the $(\{\Delta C_k, e_k\})$ parameter space and to reduce sensitivity to the initial guess, the non-linear least-squares minimization is performed with multi-start initialization, retaining the solution with the lowest residual sum of squares as an approximation to the global minimum.

Since the number of contributing defects K is generally unknown beforehand, model order selection is required. The original method employs a sequential F-test on the residual sum of squares. Starting from $K = 1$, the χ^2_K of consecutive fits is compared via the test statistic $F = [(\chi^2_K - \chi^2_{K+1})/\Delta p]/[\chi^2_{K+1}/(n - p_{K+1})]$, where n denotes the number of data points, p_K denotes the number of free parameters of the K-component model, and $\Delta p = p_{K+1} - p_K$. A new exponential component is accepted only if the corresponding F value exceeds the critical value at a chosen significance level (typically $\alpha = 0.01$); the smallest K for which no further significant improvement is observed is selected as the optimal model order. This F-test-based criterion ensures that additional components are introduced only when they are statistically justified by the data, which aims to mitigate overfitting.

The second established approach builds directly on the inverse Laplace transform of the capacitance transient. Using a Tikhonov-regularized numerical inversion such as the CONTIN algorithm,[20] the time-domain signal is decomposed into a non-negative spectral density function $f(e)$ on a discrete emission-rate grid such that

$$C(t, T) - C(V_R, T) = \int f(e)\exp(-e \cdot t)\, \mathrm{d}e. \tag{4}$$

Each defect contributing to the transient appears as a localized peak in $f(e)$ centered at its emission rate $e_k(T)$. The integral of $f(e)$ over the entire spectrum recovers the total capacitance deflection of the transient, while the contribution of an individual defect k is obtained by restricting the integration to a finite interval around its peak.[14,15] This analysis yields $\Delta C_k(T)$, from which the defect concentration is computed via Eq. (2).

Both established approaches are applied to the synthetic and the experimental data sets analyzed in this work; the corresponding results are reported in the supplementary material and used there for a qualitative and quantitative comparison with the combined two-step routine introduced below.

B. Combined Laplace transform and curve fitting

The combined evaluation routine proposed in this work is a sequential, two-step procedure whose central design choice is to decouple rate identification from amplitude determination. First, the regularized inverse Laplace transform of the measured capacitance transient $C(t, T)$ is computed at every temperature, yielding an emission-rate spectrum that reveals the K distinct rates contributing to the transient; the spectral amplitudes themselves are deliberately not interpreted at this stage. Compiling the rates $e_k(T)$ across temperatures into an Arrhenius plot provides the activation energy E_A and the capture cross section at infinite temperature σ_∞ for each defect, as shown in Fig. 1. In the second step, Eq. (3) is fitted to the measured transient in the time domain with the emission rates held fixed. Rather than reusing the Laplace-determined rates directly, they are recalculated for each temperature from the Arrhenius parameters, which enforces consistency with the identified trap characteristics and smooths out the small temperature-to-temperature scatter. The non-linear problem thereby reduces to a linear least-squares fit in the $K+1$ amplitudes $\{\Delta C_k, C(V_R, T)\}$, from which the individual defect concentrations follow via Eq. (2). The robustness of this two-step procedure relative to a joint fit of rates and amplitudes is examined further in Sec. V.

Real defects can deviate from a single discrete level in two physically distinct ways. The emission rate may follow a broadened but symmetric distribution, for instance a Gaussian distribution of activation energies around a mean value, as commonly observed in semiconductor alloys where compositional disorder broadens the defect level.[21] Alternatively, the distribution of emission rates can be asymmetric, for instance when the Poole–Frenkel effect makes the emission rate depend on the local electric field, which is inhomogeneous across the space charge region; such field-enhanced emission is present even in elemental semiconductors.[22,23] Depending on the dominant physical mechanism, the capacitance transient from Eq. (1) can be extended with the corresponding defect model. In this work, only the symmetric broadened-level model was tested for both the synthetic and experimental data; however, it did not improve the results compared to the discrete point-defect model, as discussed in Sec. V.

C. Numerical implementation

The numerical inverse Laplace transform is calculated using the CONTIN algorithm, which employs Tikhonov regularization to solve the ill-posed inverse problem.[20,24] The algorithm requires monotonically decaying transients with positive amplitudes. Since DLTS measurements commonly employ majority carrier filling pulses, the resulting transients exhibit negative amplitudes, as described in Sec. II. To meet the algorithm's requirements in this case, the measured transient is inverted and shifted to the baseline by subtracting its final value $C(t_{\mathrm{end}})$, preserving the transient shape while ensuring compatibility with CONTIN.

The emission rate spectrum boundaries are determined by the temporal characteristics of the measured transient. The upper limit is constrained by the temporal resolution of the measurement system, typically in the microsecond range, while the lower limit is set by the total acquisition duration of the transient. For the measurements presented here, with a sampling interval of $10\,\mu$s and a total acquisition time of 1 s, the emission rate range was set to $e = [0.1;\ 1 \times 10^5]$1/s with 150 discrete steps. The

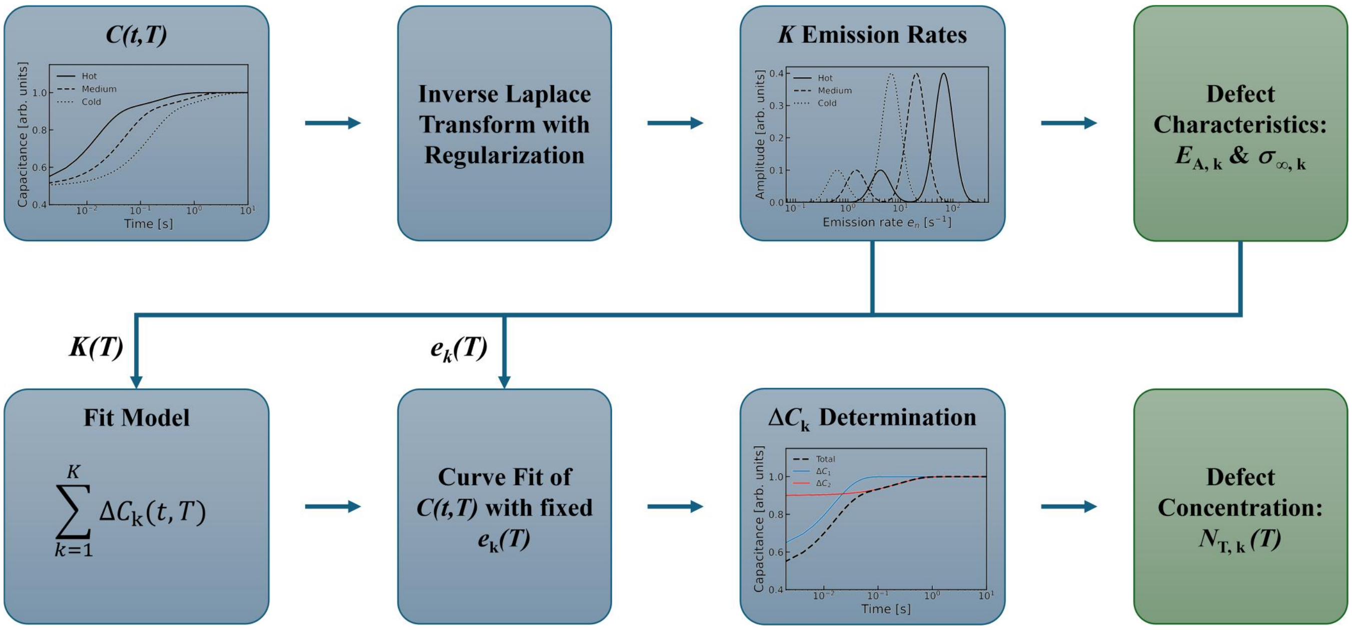

FIG. 1. Schematic representation of the proposed evaluation routine combining inverse Laplace transform for emission-rate determination with subsequent constrained curve fitting for capacitance deflection extraction.

regularization parameter lower limit was set to $R_{\mathrm{CONTIN}} = 0.2$, chosen empirically to balance noise suppression against preservation of genuine spectral features. The subsequent curve fitting is performed via non-linear least-squares minimization using the Python library lmfit.[25] With the emission rates fixed from the Arrhenius analysis, the fit converges reliably to determine the individual capacitance deflections ΔC_k for the transients at all individual temperatures.

IV. VALIDATION WITH SYNTHETIC DATA

The method is first validated using synthetic transients with known input parameters, confirming that defect characteristics and concentrations are accurately recovered under controlled conditions representative of realistic measurement scenarios. Subsequently, the routine is applied to experimental data from MOVPE-grown n-GaAs.

A. Synthetic data generation

Synthetic capacitance transients were generated according to Eq. (3) using two majority defects with characteristics similar to the *EL2* and *ELO* levels commonly observed in n-GaAs,[17] here referred to as synthetic defect 1 and 2 (*SD1*, *SD2*). The *SD1* defect was assigned an activation energy of $E_{A,1} = 711$ meV and a capture cross section of $\sigma_{\infty,1} = 1.8 \times 10^{-15}\,\mathrm{cm}^2$, while the *SD2* defect was modeled with $E_{A,2} = 658$ meV and $\sigma_{\infty,2} = 9.1 \times 10^{-15}\,\mathrm{cm}^2$. The capacitance deflections were set to $\Delta C_1 = -0.75$ pF and $\Delta C_2 = -0.05$ pF, yielding majority carrier signals with a concentration ratio of approximately 15:1. The base capacitance was modeled as temperature-independent and set to $C(V_R) = 204.5$ pF.

Transients were computed for temperatures between 300 and 480 K with a sampling rate of 100 kHz and a total acquisition time of 1 s, matching the experimental conditions described in Sec. VI A. Temperature-dependent white noise was added, with the standard deviation scaled proportionally to $\sqrt{T}$ to reflect the temperature dependence of thermal (Johnson–Nyquist) noise. At the reference temperature of 350 K, the signal-to-noise ratio was approximately 24 dB, representative of well-averaged experimental data.

The noisy transients were then processed identically to measured data: The data points were binned onto a logarithmically spaced time axis with 100 intervals by averaging within each interval, followed by the complete evaluation routine described in Sec. III.

B. Evaluation procedure and results

Figure 2 presents the complete evaluation sequence applied to the synthetic data, demonstrating each step of the method. The conventional double-boxcar evaluation [Fig. 2(a)] yields a single peak for the given rate window, with the broadening caused by overlapping emissions being barely discernible without logarithmic scaling. The dashed lines indicate the individual contributions from both defects, which merge into the observed signal. This peak broadening prevents accurate characterization using standard methods, as the overlapping emissions cannot be separated.

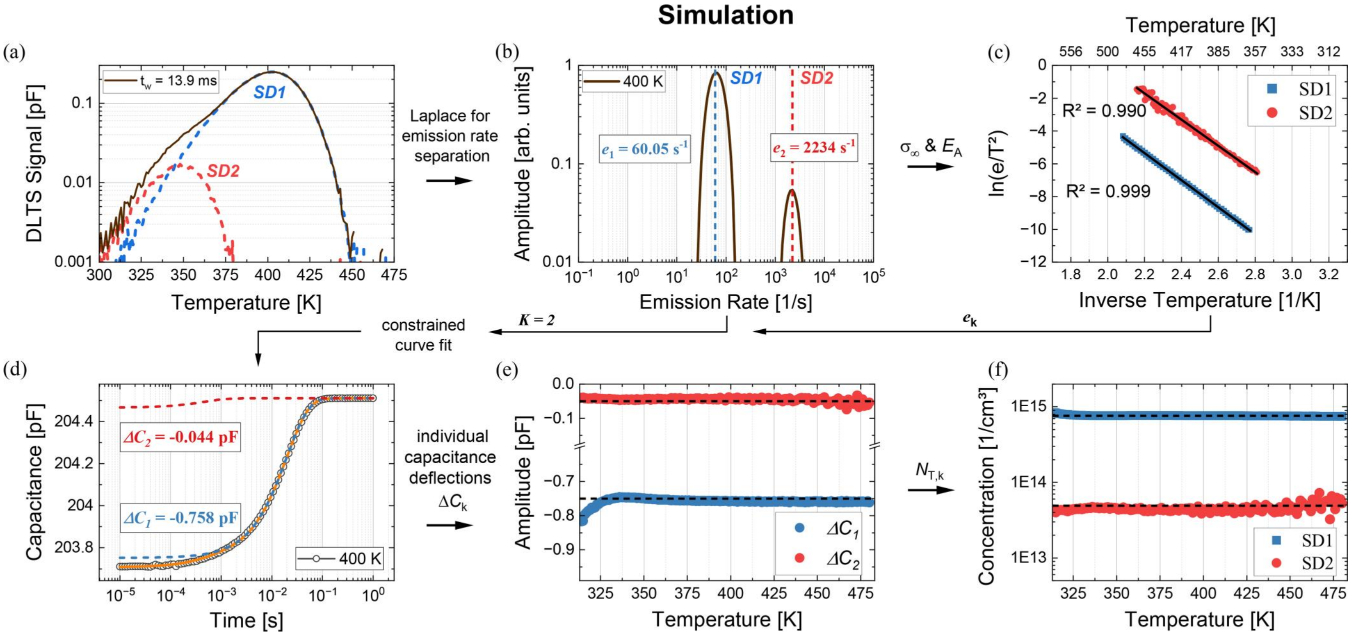

FIG. 2. Validation of the combined evaluation routine using synthetic data: (a) Conventional double-boxcar DLTS signal (brown) showing peak broadening due to overlapping defect emissions; dashed lines indicate individual contributions (blue and red). (b) Laplace transform at $T = 400$ K revealing two distinct emission rates. (c) Arrhenius plot constructed from the Laplace-determined emission rates, yielding activation energies and capture cross sections. (d) Curve fit of the capacitance transient at $T = 400$ K: synthetic data (black), total fit (orange), and individual contributions from the dominant (blue) and minor (red) defect. (e) Recovered capacitance deflections and (f) individual defect concentrations over all temperatures with dashed lines indicating modeled values.

Application of the inverse Laplace transform resolves this ambiguity. Figure 2(b) shows the spectral density function at the temperature of $T = 400$ K, clearly revealing two distinct emission rates corresponding to the two modeled defects. The emission rates extracted from the Laplace spectra are compiled in the Arrhenius plot [Fig. 2(c)]. Linear regression yields the trap characteristics for both defects, given in Table I. The recovered values of the dominant *EL2* defect match the input parameters within the uncertainty of the fit. The minor *ELO* defect shows a slightly larger deviation, particularly at lower temperatures where the dominant defect's large amplitude and slow decay impede the Laplace analysis, causing a slight overestimation of the minor defect's emission rate (see Sec. IV C). Nevertheless, the Arrhenius fit accurately captures the defect signatures for both traps.

TABLE I. Comparison of input and recovered parameters from the synthetic data validation. The columns *Laplace-integral*, *Constrained curve fit*, and *Discrete fit* list the values obtained from the proposed two-step routine, the local integration of the Laplace spectrum (with integration limits set where the spectral amplitude drops below 1% of the corresponding peak height) and the DISCRETE algorithm (with maximum model order $K_{max} = 3$), respectively. Capacitance deflections and defect concentrations are averaged over the active temperature window from 350 to 450 K.

Parameter	Input	Laplace-integral	Constrained curve fit	Discrete fit
SD1				
$E_{A,1}$ (meV)	711	712 ± 2	712 ± 2	711 ± 1
$\sigma_{\infty,1}$ (cm^2)	1.7×10^{-15}	1.8×10^{-15}	1.8×10^{-15}	1.7×10^{-15}
ΔC_1 (pF)	0.75	0.763 ± 0.002	0.757 ± 0.002	0.749 ± 0.001
$N_{T,1}$ (1/cm^3)	7.6×10^{14}	7.8×10^{14}	7.6×10^{14}	7.6×10^{14}
SD2				
$E_{A,2}$ (meV)	658	689 ± 6	689 ± 6	646 ± 3
$\sigma_{\infty,2}$ (cm^2)	9.1×10^{-15}	3.8×10^{-14}	3.8×10^{-14}	7.4×10^{-15}
ΔC_2 (pF)	0.05	0.039 ± 0.004	0.0442 ± 0.003	0.0498 ± 0.003
$N_{T,2}$ (1/cm^3)	4.9×10^{13}	$(3.9 \pm 0.4) \times 10^{13}$	$(4.4 \pm 0.3) \times 10^{13}$	$(4.9 \pm 0.2) \times 10^{13}$

With the emission rates established, the curve fitting procedure for $K = 2$ determines the individual capacitance deflections. Figure 2(d) shows the fit result at $T = 400$ K: The two-defect model (orange) accurately reproduces the synthetic transient (black), with clearly distinguishable contributions from the dominant defect (blue) and the minor defect (red). The fitted amplitudes of $\Delta C_1 = -0.758$ pF and $\Delta C_2 = -0.044$ pF closely match the input values. For the dominant defect, the relative deviation is approximately 1%. The minor defect exhibits a larger relative deviation of 12%; however, the absolute deviation of 0.006 pF is comparable to the noise level of approximately 0.004 pF, indicating that the uncertainty in ΔC_2 is noise-limited rather than method-limited.

C. Temperature dependence and active window

The recovered capacitance deflections across all temperatures are presented in Fig. 2(e). Both ΔC_1 and ΔC_2 remain constant over a broad temperature range, demonstrating that the method reliably extracts individual defect concentrations independent of temperature. However, systematic deviations for ΔC_1 appear at the lower temperature boundary of the measured range. The recovered amplitudes diverge from the true values because the emission rates become too small: the corresponding time constants exceed the 1 s acquisition window, and the transient decays only partially within the measurement interval. Consequently, the curve fit lacks sufficient information to accurately determine the capacitance deflection. The analogous effect is expected to occur at high temperatures, where the emission rates become too large. Here, the transient decays within the first few sampling points, providing insufficient information for reliable amplitude extraction. For the *SD2* (*ELO*) defect, with its higher emission rate than that of *SD1* (*EL2*), this behavior manifests at temperatures above approximately 460 K. In this region, the reduced transient information is compounded by increasing noise levels, rendering the fit results unreliable. Together, these effects define an active temperature window within which the emission rate—or equivalently, the decay time constant—falls predominantly within the measurement interval. This temperature window corresponds directly to the range where the Laplace transform successfully identifies the emission rate. Outside this window, the spectral peak either shifts beyond the accessible emission rate range or broadens due to insufficient transient information. The curve fitting method inherits this limitation. For the two modeled defects, the active temperature window where both emission rates lie within the acquisition window ranges from 350 to 450 K, which is also used for the averaged value in Table I.

D. Validation summary and comparison to established methods

The validation with synthetic data confirms that the combined evaluation routine accurately recovers both defect characteristics and individual concentrations. Table I summarizes the input and recovered parameters, alongside the corresponding values obtained from the two established methods discussed in Sec. III A. For the dominant defect, the recovered capacitance deflection deviates by less than 1% from the input value. The minor defect, with a signal amplitude 15 times smaller, shows a larger relative uncertainty of approximately 15%, consistent with the noise level of approximately 4×10^{-3} pF. This demonstrates that the method's accuracy for weak signals is noise-limited rather than method-limited: When the Laplace transform successfully identifies an emission rate, the subsequent curve fit reliably determines the corresponding capacitance deflection.

Across the active temperature window, the three evaluation routines rank consistently on the synthetic data: The DISCRETE algorithm recovers the input amplitudes most accurately, followed by the proposed curve fit routine, while the local integration of the Laplace spectrum produces the largest deviations. As visible in Table I, the integration method slightly overestimates ΔC_1 of the dominant defect *SD1* and clearly underestimates ΔC_2 of the minor defect *SD2*, whereas the constrained curve fit reproduces both amplitudes with higher accuracy.

This ranking reflects how directly each method inverts the data-generation process. DISCRETE fits Eq. (3) directly in the time domain with both emission rates and capacitance deflections as free parameters, mirroring the multi-exponential model used to generate the synthetic transients. The two other methods share the regularized inverse Laplace transform, which maps the transient onto an emission rate spectrum before any amplitude is read off. This step introduces a regularization-dependent reshaping that broadens the ideally discrete input rates into finite-width peaks. When the contributing defects exhibit strongly imbalanced signal amplitudes, as is the case here, the regularization disproportionately distorts the weaker peaks so that the introduced bias becomes non-negligible even for the ideally discrete synthetic input.[15]

The proposed two-step routine retains the rate identification from the Laplace inversion but bypasses the spectral amplitude reconstruction entirely. Once the emission rates are fixed via the Arrhenius analysis, the capacitance deflections are extracted from a constrained fit of the time-domain transient—i.e., from the same physical signal in which the defect concentration of Eq. (2) is formulated. This explains why the curve fit values remain close to the input on the synthetic data, even where the local-integration values, derived from the same Laplace spectrum, do not.

Compared to DISCRETE, the two-step routine is nonetheless marginally less accurate on the synthetic data, and this residual gap can be traced entirely to the fixed-rate constraint. For a given set of emission rates, the second step of the two-step routine reduces to the same least-squares amplitude estimation that DISCRETE solves jointly with the emission rates; its accuracy is therefore bounded by the accuracy of the rates supplied by the preceding Laplace analysis. As Table I shows, the Arrhenius-derived activation energy and capture cross section—and hence the emission rates—still carry a residual deviation from the input values, most pronounced for the minor defect *SD2*. Constraining the fit to a slightly incorrect emission rate forces a compensating bias onto the recovered amplitude. Were the emission rates recovered without error, the constrained amplitude fit and the estimate produced by DISCRETE would coincide, and both routines would recover the capacitance deflections with identical accuracy. The reliable identification and assignment of the emission rates by the Laplace analysis is therefore the decisive prerequisite of the proposed routine; once it is met, decoupling rate identification from amplitude determination carries no intrinsic accuracy penalty relative to the unconstrained multi-exponential fit.

V. METHOD DISCUSSION

The presented method combines two well-established techniques—Laplace transform analysis and least-squares curve fitting—in a sequential two-step approach. Since both underlying methods are fundamentally independent of the material system, the combined approach is applicable to any semiconductor where conventional Laplace DLTS can resolve overlapping emission rates. The method requires no material-specific assumptions beyond the standard DLTS framework and can therefore be transferred to III–V compounds, silicon, or wide-bandgap semiconductors. The key advantage of the method lies in its ability to provide reliable quantitative concentration values for individual defects whose emissions overlap in conventional DLTS.

A. Limitations and practical considerations

While the method successfully determines individual defect concentrations for overlapping emission rates, several points should be considered for practical application:

1. **Performance against established methods:** DISCRETE recovers the synthetic input parameters most accurately, operating closest to the underlying multi-exponential model. However, it treats both emission rates and capacitance deflections as free parameters, twice the variables of the constrained curve fit, and additionally requires the upper model order $K_{\max}$ to be fixed prior to the fit. When the number of contributing defects is not known a priori, $K_{\max}$ tends to be chosen conservatively high, which favors mathematically optimal but physically spurious solutions through overfitting; noise and non-ideal defect behavior amplify this tendency. These properties make DISCRETE sensitive to departures from the ideal discrete-defect picture, such as asymmetric or broadened emission-rate distributions. On the measurement data (Sec. VI), this manifests as degenerate parameter combinations and a loss of the temperature consistency that the synthetic validation suggested. The local integration of the Laplace spectrum avoids this parameter degeneracy but inherits the regularization-dependent bias of the inverse transform, already non-negligible for synthetic discrete inputs and growing on experimental data, where peak broadening and asymmetry depart further from the symmetric shape enforced by Tikhonov regularization.[15] The proposed curve fit avoids both failure modes by using each technique only where it is strongest: The Laplace inversion for its emission-rate resolution and a time-domain least-squares fit for the capacitance deflections. For the ideal case, the proposed two-step routine and DISCRETE reach comparable accuracy; the advantage of the constrained curve fit routine emerges on experimental data, where these idealized assumptions no longer hold.
2. **Signal-to-noise requirements:** The method's accuracy depends critically on the signal-to-noise ratio (SNR) of the measured transient. As demonstrated in the validation, the uncertainty for weak defect signals is noise-limited. For lower SNR, the Laplace transform may not reliably identify minor emission peaks, precluding the subsequent curve fitting analysis. The SNR threshold for reliable analysis is, however, less stringent than for single-step time-domain fits such as DISCRETE: With the emission rates fixed via the Arrhenius regression, the amplitude fit reduces to a linear problem, reducing the number of free parameters by half and limiting noise propagation compared to a simultaneous non-linear fit at each temperature.
3. **Active temperature window:** The curve fitting procedure requires that the emission rate of each defect falls predominantly within the measurement time interval. For the acquisition parameters used here ($t = [5 \times 10^{-5};\ 1]$ s), this constrains the accessible emission rates to approximately $e = [0.5;\ 4 \times 10^4]$ 1 s. Outside this window, the transient either decays too slowly (low temperatures) or too rapidly (high temperatures) to provide sufficient information for reliable amplitude determination. This limitation is inherent to the Laplace analysis and also constrains the curve fitting method.
4. **Discrete defect levels:** The implementation presented here assumes discrete point defects with well-defined emission rates. Given the broadened DLTS peak observed in the synthetic data (and also in the experimental data in Sec. VI), an alternative model incorporating a single broad energy level for *SD1* (*EL2*) (Gaussian-distributed) without the *SD2* (*ELO*) level was also tested.[21] In this approach, the variance of the Gaussian distribution was introduced as an additional fit parameter alongside the capacitance amplitude. However, this single-defect model with energy broadening failed to accurately reproduce the measured transient, yielding systematic S-shaped residuals characteristic of an inadequate model. Only the two-defect model with $K = 2$ discrete emission rates produced non-systematic residuals, confirming that the observed peak broadening originates from two overlapping point defects rather than from energy level broadening of a single defect. However, for other defect signals, the broad energy distribution might be the more accurate fit model.
5. **Quality assessment:** The residual analysis described above exemplifies the built-in quality criteria the method provides for assessing the reliability of results. Residual analysis of the curve fit reveals whether a single-, multi-, or broad-defect model (or a combination of those) adequately describes the measured transient: Systematic residual patterns indicate missing emission components, while random residuals confirm that all significant contributions are captured. Furthermore, the temperature independence of the recovered concentrations within the active window [cf. Figs. 2(e) and 3(e)] serves as a consistency check. Deviations from constant concentration values—beyond the expected behavior at the window boundaries—would indicate either fitting artifacts or non-ideal defect behavior.
6. **Maximum number of emission rates:** The method was also applied to simulations (and measurements) where $K = 3$ emission rates were present within the same active temperature window, yielding reliable results. In principle, no inherent limitation restricts the number of defects that can be analyzed: The curve fitting approach assigns an individual amplitude to each emission rate identified by the Laplace transform. The practical

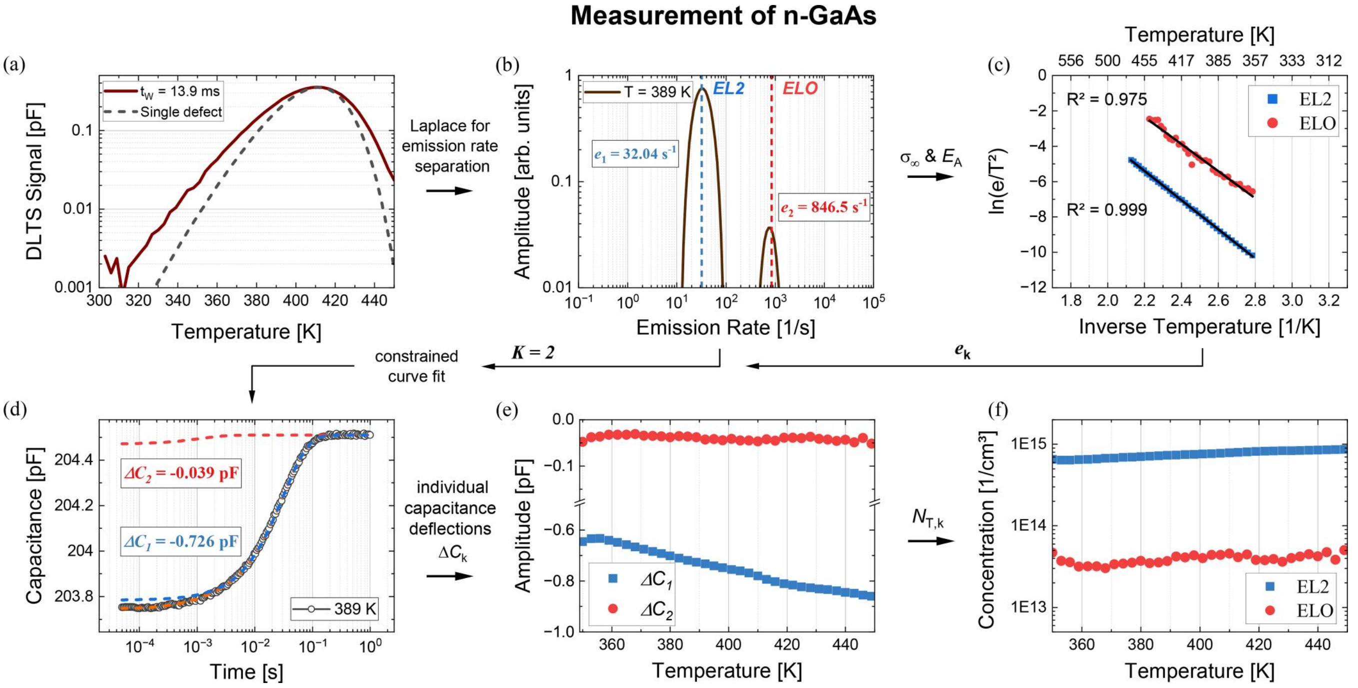

FIG. 3. Application of the combined evaluation routine to MOVPE-grown n-GaAs. (a) Conventional double-boxcar DLTS signal showing a broadened peak (brown) indicative of overlapping defect emissions. The expected signal of an ideal defect signal (dashed) is additionally shown for comparison. (b) Laplace transform of the transient at the temperature of $T = 389$ K revealing two distinct emission rates corresponding to *EL2* and *ELO*. (c) Arrhenius plot yielding activation energies and capture cross sections for both defects. (d) Curve fit of the capacitance transient at $T = 389$ K: measured data (black), total fit (orange), and individual contributions from *EL2* (blue) and *ELO* (red). (e) Recovered capacitance deflections and (f) defect concentrations determined across all temperatures.

limit is therefore set by the resolution capability of the Laplace analysis rather than by the curve fitting procedure itself. Furthermore, the method naturally extends to systems where defects exhibit well-separated active temperature windows. In this case, the fit model is constructed at each temperature using only those K emission rates that fall within the accessible measurement interval, while defects outside their active window are excluded from the model since their transient contribution is negligible. This temperature-adaptive model construction allows the method to be applied to a broad range of defect systems regardless of the total number of defects present in the material.

VI. APPLICATION TO MOVPE-GROWN GaAs

Having validated the combined evaluation routine with synthetic data, the method is applied to experimental data from a measurement on n-type GaAs to demonstrate its practical utility for quantitative defect analysis.

A. Experimental details

A n-GaAs Schottky diode was fabricated from material grown by metalorganic vapor-phase epitaxy at a growth rate of 94 μm/h in an AIXTRON Crius Close Coupled Showerhead reactor; the growth procedure is described in detail elsewhere.[18] The n-GaAs layer with a doping concentration of $N_D \approx 2 \times 10^{16}/\text{cm}^3$ was grown on a 4-in. n-GaAs (100) wafer with a 6° offcut toward the ⟨111⟩B direction. Nickel contacts with a circular area of 1.02 mm^2 were electroplated onto the bare GaAs surface to form a Schottky junction.[26] Capacitance transients were recorded using a Zurich Instruments MFIA impedance analyzer, controlled by a Python-based measurement routine.[27] The measurement sequence consisted of pulsing the diode from −3.5 to −2 V for 50 ms, followed by transient acquisition for 1 s at a sampling rate of 100 kHz. The AC test signal was set to 75 mV at 700 kHz, and each transient was averaged over 1350 repetitions. Temperature was controlled using a Lakeshore 332 controller with the sample mounted in a cryostat under high vacuum ($<10^{-6}$ mbar). Measurements were performed over a temperature range from 300 to 450 K in 3 K steps with a stabilization criterion of less than 0.1 K for 1 min. Prior to the DLTS measurements, capacitance–voltage profiling at the same AC settings determined the doping concentration and built-in voltage.[13] The acquired transients were binned to a logarithmically spaced time axis with 100 points by averaging within each interval for subsequent analysis. For the binned transients, the signal amplitude was taken as the sum of the capacitance amplitudes of all fitted components, while the measurement noise was estimated from the first differences of the fit residuals in the decayed portion of the transient, with the noise standard deviation given by the standard deviation of these differences divided by $\sqrt{2}$. Expressed as the amplitude ratio $SNR = A/\sigma$, this yielded a median signal-to-noise ratio of ∼3700 across the measured temperature range.

B. Defect characterization

Figure 3 presents the complete evaluation sequence applied to the experimental data, following the same procedure validated in Sec. IV. The conventional double-boxcar evaluation [Fig. 3(a)] reveals a single dominant peak. However, closer inspection shows asymmetric broadening on both sides of the peak compared to the theoretical response of an ideal point defect, suggesting the presence of overlapping emission processes that the double-boxcar method cannot resolve. While the low-temperature shoulder is clearly visible, the high-temperature side cannot be fully resolved as the maximum measurement temperature is technically limited to 450 K.

Application of the inverse Laplace transform confirms this hypothesis. Figure 3(b) shows the spectral density function at the temperature of $T = 389\,K$, clearly revealing two distinct emission rates. The Arrhenius analysis of both signals is presented in Fig. 3(c). The dominant emission yields trap characteristics of $E_{A,1} = 711\,\text{meV}(\pm 2)$ and $\sigma_{\infty,1} = 1.7 \times 10^{-15}\,\text{cm}^2$, identifying the defect as the midgap level *EL2*, commonly observed in n-GaAs.[28] The second emission exhibits an activation energy of $E_{A,2} = 658\,\text{meV}(\pm 20)$, approximately 50 meV lower but the capture cross section of $\sigma_{\infty,2} = 9.1 \times 10^{-15}\,\text{cm}^2$ is nearly one order of magnitude higher. These characteristics match the oxygen-related *ELO* defect, first identified by Lagowski *et al.*[17]

C. Determination of defect concentration

With both defects identified, the curve fitting procedure determines their individual capacitance deflections. Figure 3(d) shows the fit result at $T = 389$ K: The two-defect model (orange) accurately reproduces the measured transient (black), with clearly distinguishable contributions from *EL2* (blue) and *ELO* (red).

The curve fitting procedure was applied to all transients within the active temperature window from 350 to 450 K, where the emission rates of both defects fall within the accessible measurement interval. The resulting capacitance deflections are presented in Fig. 3(e). The amplitude ΔC_2 of *ELO* remains constant over the entire temperature range. In contrast, ΔC_1 of the *EL2* exhibits a continuous increase with increasing temperature. This phenomenon is frequently observed in DLTS, where the peak amplitude increases with temperature.[29] Since this behavior is present in the raw measurement data, it manifests as an increasing ΔC_1 in the curve fit. Notably, ΔC_2 remains unaffected by this effect. The physical origin of the temperature-dependent amplitude requires further investigation but is not attributed to the evaluation method itself. The defect concentrations calculated from the fitted amplitudes according to Eq. (2) are shown in Fig. 3(f). The mean concentrations are $N_{T,1} = (7.5 \pm 0.7) \times 10^{14}/\text{cm}^3$ for *EL2* and $N_{T,2} = (3.9 \pm 0.5) \times 10^{13}/\text{cm}^3$ for *ELO*. The slight temperature dependence of ΔC_1 propagates into a corresponding drift in the *EL2* concentration, while the *ELO* concentration remains stable. Table II summarizes the complete defect parameters. The *ELO* concentration is approximately 20 times lower than that of *EL2*. However, the capture cross section of *ELO* is roughly one order of magnitude higher. Since the Shockley–Read–Hall recombination rate scales with the product of concentration and capture cross section, both defects contribute comparably to carrier recombination dynamics despite their vastly different concentrations.[13]

The two established methods were applied to the same experimental data for comparison; the corresponding values are compiled in Table II and documented in detail in the supplementary material. Local integration of the Laplace spectrum reproduces the dominant *EL2* amplitude and its temperature drift, but systematically underestimates the minor *ELO* amplitude relative to the constrained curve fit. This underestimation is consistent with the synthetic validation (Sec. IV D) and reflects the tendency of Tikhonov-regularized inversions to narrow minor peaks coexisting with a much larger dominant peak;[15] operating in the time domain, the constrained curve fit is not subject to this shape distortion and provides the more reliable *ELO* estimate. The DISCRETE algorithm, accurate on synthetic data, encounters

TABLE II. Defect parameters determined for MOVPE-grown n-GaAs by the different evaluation routines: The constrained curve fit alongside the local Laplace-integral and the DISCRETE results ($K_{max} = 2$). Capacitance deflections and defect concentrations are averaged over the active temperature window from 350 to 450 K.

Parameter	Constrained curve fit	Laplace-integral	Discrete fit
EL2			
E_A (meV)	711 ± 2	711 ± 2	684 ± 2
σ_∞ (cm^2)	1.7×10^{-15}	1.7×10^{-15}	7.5×10^{-16}
ΔC (pF)	0.750 ± 0.075	0.772 ± 0.068	0.711 ± 0.061
N_T ($1/\text{cm}^3$)	$(7.5 \pm 0.7) \times 10^{14}$	$(7.8 \pm 0.7) \times 10^{14}$	$(7.1 \pm 0.6) \times 10^{14}$
ELO			
E_A (meV)	658 ± 20	658 ± 20	437 ± 24
σ_∞ (cm^2)	9.1×10^{-15}	9.1×10^{-15}	5.7×10^{-18}
ΔC (pF)	0.040 ± 0.005	0.0265 ± 0.0034	0.0711 ± 0.0510
N_T ($1/\text{cm}^3$)	$(3.9 \pm 0.5) \times 10^{13}$	$(2.6 \pm 0.3) \times 10^{13}$	$(7.0 \pm 5.0) \times 10^{13}$

distinct failure modes on the experimental transients. With $K_{\max} = 2$, the dominant *EL2* component is recovered mostly consistently: The extracted emission rates agree with the Laplace results and the literature defect characteristics.[17] For the minor *ELO* component, by contrast, the emission rates deviate systematically from both Laplace and literature values, and since amplitude and rate enter the non-linear fit as coupled parameters, the fitted pair (ΔC, e) is less reliable. Raising $K_{\max}$ to 3 produces overfitting in almost all transients: A spurious third component appears between the physical *EL2* and *ELO* rates and splits the defect signal across three exponentials so that neither the emission rates nor the amplitudes can be assigned unambiguously to individual defects.

The constrained curve fit avoids both failure modes by fixing the emission rates from the globally consistent Arrhenius analysis prior to amplitude determination, preserving the assignment of capacitance deflections to individual defects across the full active window. The susceptibility of DISCRETE to overfitting compromises the recovery of the weak *ELO* signature even at $K_{\max} = 2$, which illustrates the practical benefit of decoupling rate identification from amplitude determination.

D. Practical implications

Recent studies on defect formation in n-GaAs grown by MOVPE have identified a strong increase in *EL2* concentration at elevated growth rates.[30] The *EL2* defect, associated with the arsenic antisite[28] (As_{Ga}), was identified as the performance-limiting defect in these materials, since increasing the growth temperature reduced the *EL2* concentration and improved device performance.[30] The evaluation method presented here enables a more comprehensive assessment. To investigate the growth rate dependence of both *EL2* and *ELO*, a reference sample grown at 7 μm/h was measured in addition to the high-rate sample (94 μm/h) discussed above. Both defects were detected at the lower growth rate, with concentrations of approximately $(1.2 \pm 0.2) \times 10^{14}/\text{cm}^3$ for *EL2* and $(1.0 \pm 0.2) \times 10^{13}/\text{cm}^3$ for *ELO*.

The results reveal that both defects exhibit significantly higher concentrations at elevated growth rates: *EL2* increases by a factor of approximately 6, while *ELO* increases by nearly a factor of 4. This finding suggests that not only the arsenic antisite-related *EL2* but also the oxygen-related *ELO* defect plays a critical role in determining material quality at high growth rates. Given that *ELO* exhibits a noticeably higher capture cross section than *EL2*, its contribution to Shockley–Read–Hall recombination may be comparable despite its lower absolute concentration.

These observations highlight the importance of quantifying individual defect concentrations in overlapping DLTS signals. Optimization strategies focusing solely on *EL2* reduction may prove insufficient if *ELO*—previously undetected due to signal overlap—remains a significant recombination pathway. The method presented here provides the necessary tool to identify and quantify such small signal defects, enabling more targeted approaches to material optimization for high-rate MOVPE growth.

VII. CONCLUSION

This work introduces a two-step evaluation method that combines Laplace transform analysis with constrained curve fitting to determine individual defect concentrations from overlapping DLTS signals. The established alternatives of local integration of the Laplace spectrum and unconstrained multi-exponential fitting (DISCRETE) suffer from regularization-dependent amplitude bias and susceptibility to overfitting or nonphysical solutions, respectively. The proposed routine avoids both limitations by decoupling rate identification from amplitude determination: Emission rates determined from Arrhenius analysis of the Laplace spectra are held fixed in a constrained least-squares fit, reducing the problem to a robust linear determination of capacitance deflections in the time domain.

Validation with synthetic data demonstrates that the method recovers capacitance amplitudes with noise-limited uncertainty. Application to n-GaAs grown by high-rate MOVPE reveals an *ELO* concentration of $(3.9 \pm 0.5) \times 10^{13}/\text{cm}^3$—20 times lower than the dominant *EL2* defect—a quantity inaccessible through conventional double-boxcar DLTS evaluation. Comparison with a sample grown at low rate demonstrates that both defects increase in concentration at elevated growth rates, suggesting that the oxygen-related *ELO* may be equally relevant as *EL2* for material optimization in high-rate MOVPE.

This finding illustrates the broader value of the method: By quantifying individual defect concentrations, it enables identification of the dominant recombination pathways and targeted optimization of growth conditions. The approach is material-independent and applicable wherever Laplace DLTS can resolve overlapping emission rates, providing a practical tool for defect analysis in semiconductor development for a variety of applications.

SUPPLEMENTARY MATERIAL

See the supplementary material for a detailed description of the two established reference methods used for comparison in this work and their application to both the synthetic dataset of Sec. IV and the experimental n-GaAs measurement of Sec. VI. Section S1 documents the local integration of the Laplace spectrum. Section S2 details the implementation of Provencher's DISCRETE algorithm employed here.

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AUTHOR DECLARATIONS

Conflict of Interest

The authors have no conflicts to disclose.

Author Contributions

Christoph Klein: Conceptualization (lead); Data curation (lead); Investigation (lead); Methodology (lead); Software (lead); Validation (lead); Visualization (lead); Writing – original draft (lead). **Carmine Pellegrino:** Conceptualization (equal); Methodology (equal); Validation (equal); Writing – review & editing (equal). **David Lackner:** Conceptualization (equal); Funding acquisition (lead); Methodology (equal); Project administration (lead); Validation (equal); Writing – review & editing (equal).

DATA AVAILABILITY

The data that support the findings of this study are available from the corresponding author upon reasonable request.

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